Independent retained pre-exec gate software audit

Result: retained fixed software execution is supported by its source, admission and native receipts. All27 proposal files and364 compiler/header/tool bindings hash-match. Admission joins match the exact v2 control proposal and prospective root review. No compile, gate/helper rerun, Docker action or tested payload execution was performed by this review.

The root run compiled pristine gate.c and a separate control translation unit with exact static compiler commands, -O2,-Wall,-Wextra,-Werror,-static and x86-64 baseline flags. The helper includes the same gate source with main renamed and only the mountinfo open redirected. The fixed fixture directory is compile-time pinned. The runner checks proposal/compiler identities before and after execution and uses a restricted child environment.

All45 assertions passed in five groups: nonce7; mountinfo9; exact-value parsing including descriptor open/close11; bounded-reader12; host/fixed-case rejection6. The counts derive independently from the control source loops. The pristine gate separately exited126 with 'qualification gate: PID/UID/GID boundary'. The controls deliberately refuse hostPID1 and do not reach the production kernel/exec path. They do not simulate successful kernel enforcement.

Binary digests match retained receipts: gate65f521d50818f593f5bc844eb26115bffd9866d4059a9ab5efee46df67d00377; controls1976faae3ceabd51f17125f356552993df5acca89c3e50447ac749101567d102. Independent byte parsing confirms both are ELF64 little-endian x86-64 EXEC with no interpreter or dynamic segment, agreeing with retained readelf output. No executable was run to obtain this independent readback.

The link map loads private musl Scrt1/crti/crtn and libc.a, with GCC11 crtbeginS/crtendS. This is the private-musl C gate lineage, distinct from the Rust target's bundled libc selected by the earlier ecbench build. Current compiler/header/tool hashes match the bound compilation inputs; this is custody consistency, not a fresh reproducible-build demonstration or complete historical header-access trace.

Disassembly of main follows the source identity, cgroup mapping, exact controller reads, bounded receipt write and two execve branches. No fork/clone/thread-creation call is present in the gate source or observed main call sequence. Runtime tm_clone registration symbols are CRT bookkeeping, not process cloning. This source/disassembly inspection is not an exhaustive formal machine-code proof of all transitive library behavior. The source separately requires PID1/UID1000, private read-only cgroup2 mapping and pids.max/current1 before tested exec.

Verdict scope: fixed software controls and retained binary provenance validated; actual container kernel qualification remains unperformed here. The qualifier must bind this exact gate digest, stage the immutable read-only executable and validate its live nonce/case/pre-exec receipt in each active case. Reviewer previously authored the private musl compiler prerequisite and reviewed qualifier source; this involvement is disclosed. Root produced these execution receipts. No scientific timing or cost ratio is inferred.